

A Shared operator view

$$\mathbf{X}_b = \mathbf{X} \mathbf{A}_b^T$$

linear spectral operator



$$\mathbf{A}_b \in \mathbb{R}^{p \times p}$$

identity always included



fold-local branches
SNV / MSC / ASLS

B AOM-PLS

$$\mathbf{S} = \mathbf{X}^T \mathbf{Y}$$

cross-covariance



$$\mathbf{S}_b = \mathbf{A}_b \mathbf{S}$$

because $(\mathbf{X} \mathbf{A}_b^T)^T \mathbf{Y} = \mathbf{A}_b \mathbf{X}^T \mathbf{Y}$



$$\mathbf{B} = \mathbf{Z}(\mathbf{P}^T \mathbf{Z})^+ \mathbf{Q}^T$$

original-space coefficients

C AOM-Ridge

$$\mathbf{K}_b = \mathbf{X}_c \mathbf{A}_b^T \mathbf{A}_b \mathbf{X}_c^T$$

operator geometry



$$\mathbf{C} = (\mathbf{K}_b + \alpha \mathbf{I})^{-1} \mathbf{Y}_c$$

dual ridge solve



$$\boldsymbol{\beta}_b = \mathbf{A}_b^T \mathbf{A}_b \mathbf{X}_c^T \mathbf{C}$$

original-space coefficients